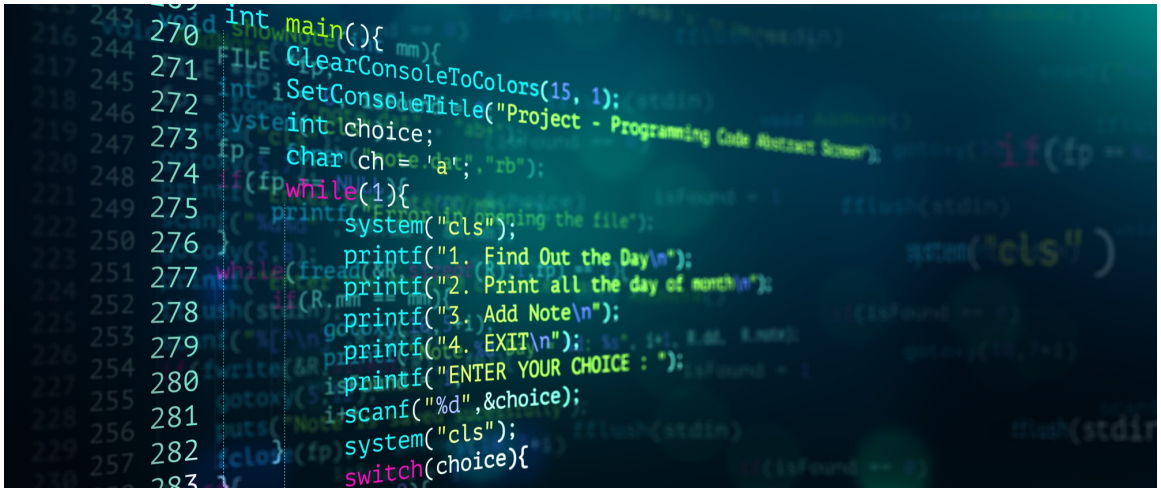

C - exercise book

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C - exercise book

Here you will find a collection of programming exercises in the C language. The inspiration was to compile tasks created during teaching classes carried out at the Faculty of Mathematics and Computer Science of the University of Warmia and Mazury in Olsztyn.

Some of the exercises have been machine-translated into English. In case of any ambiguities, please get in touch via email.

Part I

Fundamentals of the C Language

1 Input and Output Operations

1. Write a programme that asks the user to enter a single integer, then displays it on the screen.
2. Create a programme that reads two floating-point numbers from the user and prints their difference.
3. Design an application that asks the user for their year of birth, then prints the previous year (one less).
4. Write a programme that reads three integers from the user and prints their average as a floating-point value.
5. Create a programme that asks the user to enter two letters (characters), then prints them in reverse order.
6. Write a programme that asks the user to enter a single floating-point number, then doubles its value and prints the result.
7. Create a programme that reads a floating-point number from the user representing an amount in dollars, then prints how much that is in euros, using a fixed exchange rate (e.g. 1 dollar = 0.85 euros).
8. Write a programme that displays the following text on the screen: `This is a quote: "I often use the C language."`. Ensure that the quotation marks are correctly displayed as part of the string.
9. Create a programme that prints a Windows folder path to standard output, e.g. `C:\Program Files\MyApplication\` (hardcoded, without retrieving anything from the system) or `C:\\Program Files\\MyApplication\\`.
10. Design a programme that demonstrates how to print the following text on the screen: `Special characters: \t (tab), \n (newline), % (percent), \\ (backslash).`
11. Write a programme that reads two rational numbers from standard input representing the lengths of the legs of a right-angled triangle. Then calculate and display the length of the hypotenuse.
12. Write a programme that reads an integer from standard input and prints its absolute value to standard output.
13. Write a programme that reads a floating-point number from standard input and prints its absolute value to standard output.
14. Find an example and display on standard output a case where the addition of floating-point numbers is not associative.

2 Arithmetic Operators

1. Write a programme that calculates the remainder of dividing the sum of two integers by a third integer.
2. Create a programme that calculates the difference of squares of two given integers.
3. Write a programme that calculates the product of the difference of two integers and a third integer.
4. Design a programme that calculates the geometric mean of the absolute values of three given integers.
5. Write a programme that calculates the square of the sum of two given integers.
6. Create a programme that calculates the sum of squares of three given integers.
7. Write a programme that calculates how many times one given integer fits into another given integer.
8. Design a programme that calculates the square of the difference of two given integers.
9. Write a programme that calculates the quotient of the sum of two integers divided by their difference.
10. Create a programme that calculates the sum of three consecutive integers, starting from a given integer.
11. Find the arithmetic mean of three floating-point numbers.
12. Calculate the value of the expression $\frac{1}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}}$, where a, b, c are non-zero.
13. Calculate the value of e^x for small values of x, e.g. $|x| < 0.01$, approximating e^x as $1 + x$, without using exponential functions.
14. Calculate the area of a triangle with sides a, b and the angle C between them in degrees, using the formula $0.5 \cdot a \cdot b \cdot \sin C$, assuming $\sin C \approx C$ for small angles in radians.
15. Calculate $\sqrt[3]{x}$ for small values of x, e.g. $|x| < 0.01$, approximating $\sqrt[3]{x} \approx 1 + \frac{1}{3}x$, without using any root functions.

3 Conditional statements, conditional operator

In the following exercises, use `if`, `if else`, or `switch case` as the conditional statement. The conditional operator refers to `?`.

1. Write a programme that asks the user to enter an integer. The programme should display whether the entered number is positive, negative, or equal to zero.
2. Write a programme that takes two integers from the user and displays the larger of the two.
3. Write a programme that asks the user to enter a grade on a scale from 1 to 5. The programme should display a description of the grade: fail (1), poor (2), satisfactory (3), good (4), very good (5). For a number outside the range, the programme should display an error message.
4. Write a programme that asks the user to enter three distinct integers and displays the smallest of them.
5. Write a programme that asks the user for a year and checks whether the given year is a leap year. A leap year is one that is divisible by 4 but not divisible by 100, unless it is also divisible by 400.
6. Write a programme that takes two integers from the user and displays whether the sum of both numbers is even or odd.
7. Write a programme that takes three floating-point numbers `a, b, c` from the user. Treat them as the coefficients of a quadratic equation $ax^2 + bx + c = 0$. Print all solution variants of this equation to standard output.
8. Write a programme that takes two integers as input from the user and uses the conditional operator to find and display the larger of the two.
9. Write a programme that takes three integers as input from the user and uses the conditional operator to find and display the smallest of them.
10. Using the conditional operator `?`, write a programme that takes a single integer from the user and displays "even" or "odd" depending on the value of the number.
11. Create a programme that asks the user to enter two floating-point numbers and uses the conditional operator to display which of them is larger, or whether they are equal to two decimal places.

12. Using the conditional operator, write a programme that asks the user to enter a grade on a scale from 0 to 100 and displays “Pass” if the grade is greater than or equal to 51, or “Fail” otherwise.

13. Write a programme that takes a year as input from the user and uses the conditional operator to check whether it is a leap year. The programme should display “Leap year” or “Not a leap year” depending on the result.

4 Bitwise operators

1. Swap the values of two integer variables without using an additional variable.
2. Check whether an integer is even or odd using bitwise operators.
3. Determine the value of the bit at a specified position in an integer.
4. Clear the value of the bit at a specified position in an integer.
5. Invert the values of all bits in an integer.
6. Shift the bits of an integer a specified number of positions to the left.
7. Shift the bits of an integer a specified number of positions to the right.
8. Determine the XOR of two integers.

Bibliography and Other Problem Sets

Krzaczkowski, Jacek. 2011. *Zadania z Programowania w Języku c/c++, Cz. I.* Instytut Informatyki UMCS Lublin.